MLT- Lab 3

Title:

Feature Selection Methods Comparison: Filter, Wrapper, and Embedded Approaches with PCA Analysis

Dataset Descriptiom:

- **Size & Structure:** 1000 samples, 20 features (columns), 3 target classes.
- **Attributes:** All 20 columns are named 'Feature_1' to 'Feature_20', representing synthetic features generated for multiclass classification.
- **Data Types:** All features are of type float64, representing continuous numerical values.
- **Distribution:** The features follow a standardized distribution with varying means and standard deviations. The dataset was created using make_classification with 15 informative features and 5 redundant features, resulting in features that contribute differently to class separation. The target variable is balanced across 3 classes (Class 0, Class 1, Class 2).
- **Train-Test Split:** 70% training data (700 samples) and 30% testing data (300 samples), stratified by class to maintain class distribution.

Pseudo code:

```
ALGORITHM: Feature Selection and Classification with PCA Comparison

INPUT: Dataset X (features), y (target labels), k=10

STEP 1: Data Preparation
   Load multiclass classification dataset
   Split data into X_train, X_test, y_train, y_test (70-30 split)

STEP 2: Feature Selection Methods

   METHOD A - FILTER (ANOVA F-test):
      FOR each feature in X_train:
         Calculate F-statistic between feature and target
         Calculate p-value for statistical significance
      ENDFOR
      Rank features by F-score
      Select top k=10 features
      Transform X_train_filter and X_test_filter
```

```
    METHOD B - WRAPPER (Recursive Feature Elimination):
      Initialize base estimator (Logistic Regression)
      SET n_features_to_select = 10
      REPEAT:
        Train model on current feature set
        Rank features by importance/coefficients
        Remove least important feature
      UNTIL n_features = 10
      Transform X_train_wrapper and X_test_wrapper

    METHOD C - EMBEDDED (Random Forest Feature Importance):
      Train RandomForestClassifier on X_train
      Extract feature_importances_ from trained model
      Rank features by importance scores
      Select top k=10 features with highest importance
      Transform X_train_embedded and X_test_embedded

STEP 3: Model Training and Evaluation (Without PCA)
  FOR each method in [Filter, Wrapper, Embedded]:
    Scale features using StandardScaler

    # K-Nearest Neighbors
    Initialize KNN with k=10
    Train KNN on scaled training data
    Predict on test data
    Calculate accuracy score
    Perform 5-fold cross-validation
    Store results

    # Logistic Regression
    Initialize LogisticRegression with max_iter=1000
    Train model on scaled training data
    Predict on test data
    Calculate accuracy score
    Perform 5-fold cross-validation
    Store results
  ENDFOR

STEP 4: Apply PCA Transformation
  FOR each method in [Filter, Wrapper, Embedded]:
```

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    Scale features using StandardScaler
    Initialize PCA with n_components=10
    Fit PCA on scaled training data
    Transform training and testing data
    Calculate explained variance ratio
    Store PCA-transformed datasets
  ENDFOR

STEP 5: Model Training and Evaluation (With PCA)
  FOR each method in [Filter, Wrapper, Embedded]:
    # K-Nearest Neighbors
    Initialize KNN with k=10
    Train KNN on PCA-transformed training data
    Predict on PCA-transformed test data
    Calculate accuracy score
    Perform 5-fold cross-validation
    Store results

    # Logistic Regression
    Initialize LogisticRegression with max_iter=1000
    Train model on PCA-transformed training data
    Predict on PCA-transformed test data
    Calculate accuracy score
    Perform 5-fold cross-validation
    Store results
  ENDFOR

STEP 6: Comparison and Consistency Analysis
  FOR each method in [Filter, Wrapper, Embedded]:
    Compare accuracies: Without PCA vs With PCA
    Calculate mean accuracy across both models
    Calculate standard deviation (consistency metric)
    Determine performance difference

 END

  Identify best performing configuration
  Identify most consistent method (lowest std deviation)
  Analyze PCA impact on performance
```

Result analysis:

ACCURACY COMPARISON TABLE

Feature Selection	Model	Without PCA	With PCA	Difference
Filter	KNN	0.686667	0.686667	0.0
Filter	LR	0.636667	0.636667	0.0
Wrapper	KNN	0.740000	0.740000	0.0
Wrapper	LR	0.696667	0.696667	0.0
Embedded	KNN	0.683333	0.683333	0.0
Embedded	LR	0.640000	0.640000	0.0

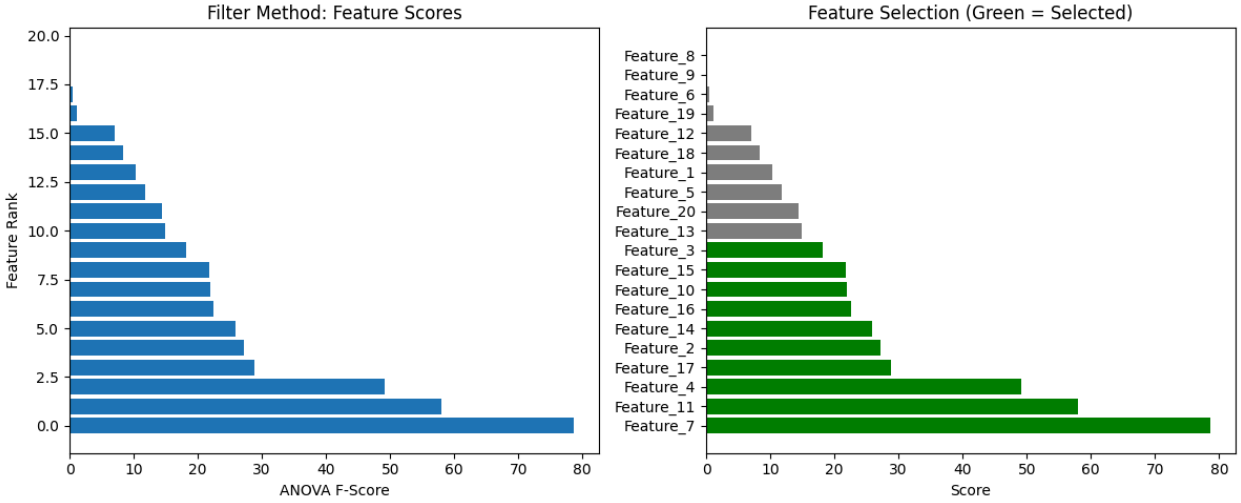
CONSISTENCY ANALYSIS

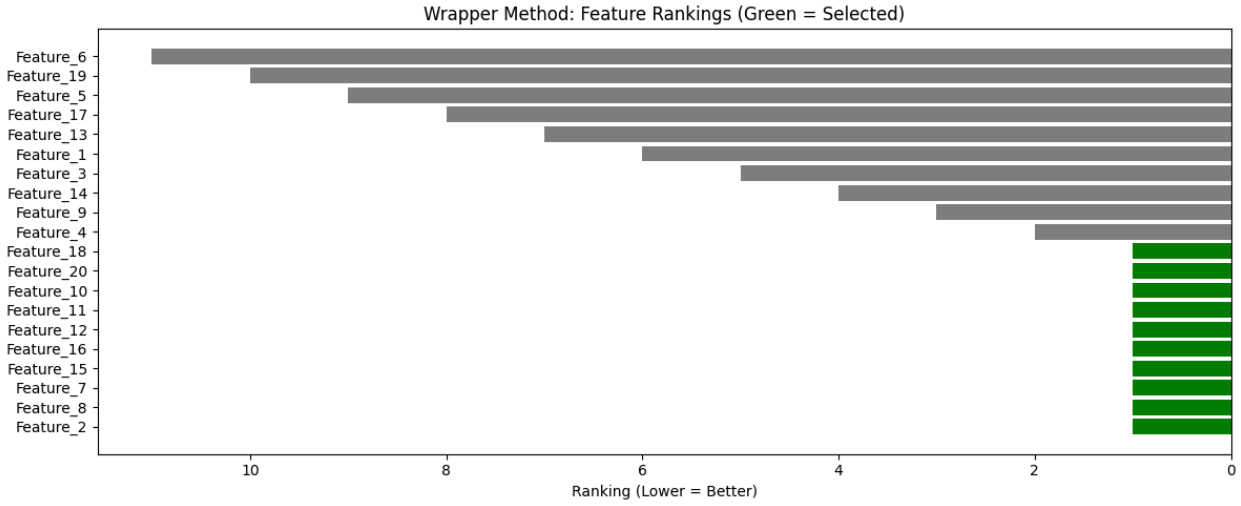
Method	Without		With		More Consistent
	PCA Mean	PCA Std	PCA Mean	PCA Std	
Filter	0.661667	0.025000	0.661667	0.025000	Without PCA
Wrapper	0.718333	0.021667	0.718333	0.021667	Without PCA
Embedded	0.661667	0.021667	0.661667	0.021667	Without PCA

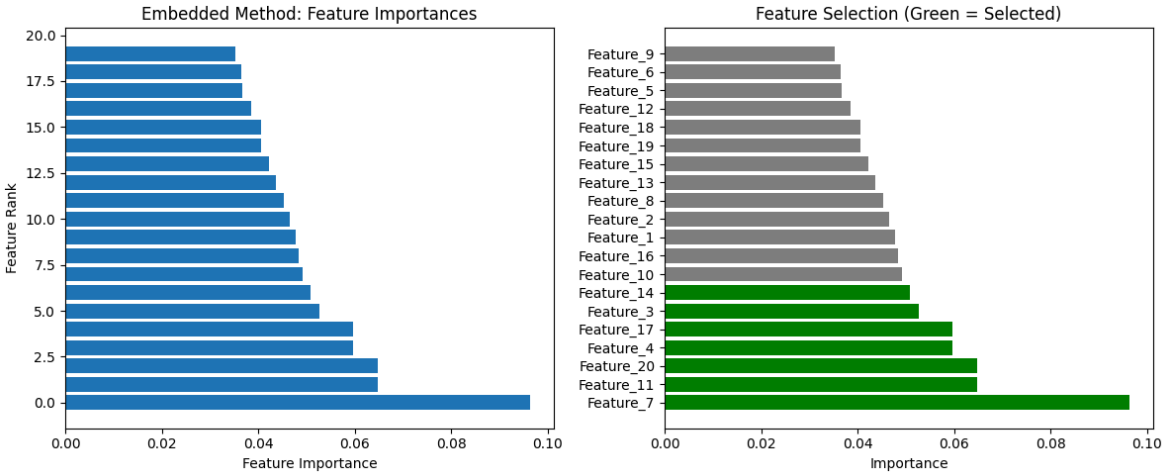
Observation:

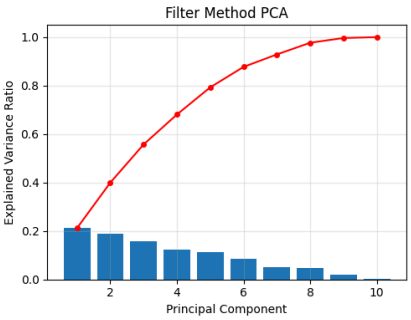
Based on these results, the **Wrapper method** (specifically RFE) is the recommended feature selection technique for this dataset, paired with a **K-Nearest Neighbors (KNN)** classifier. The application of PCA did not yield any performance benefits, indicating that the original features (after selection by the Wrapper method) are sufficient or that the data structure is not significantly simplified by PCA in a way that helps these models. Therefore, using the original features selected by the Wrapper method without an additional PCA step would be the most effective and efficient approach for this problem.

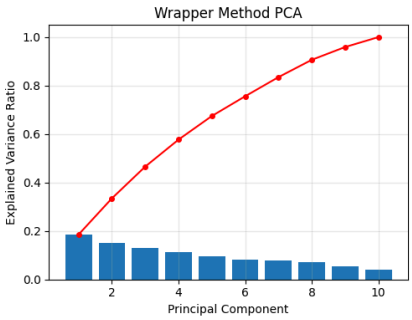
Additional Data:

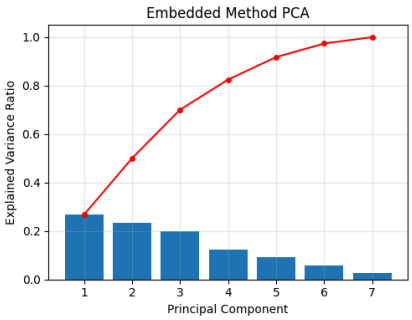


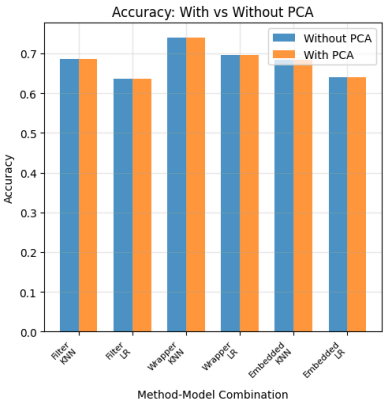


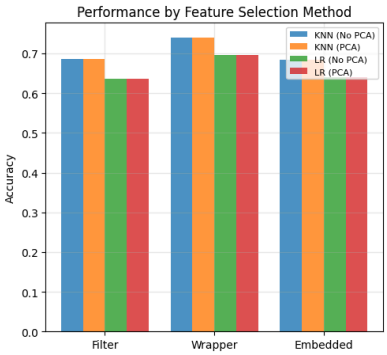


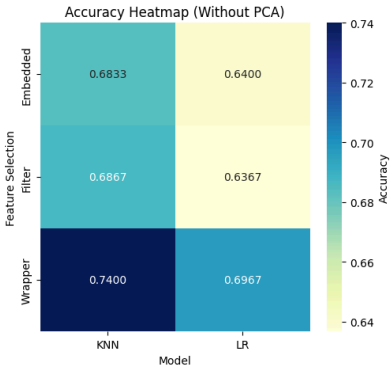

Filter Method PCA
Explained Variance Ratio
Principal Component

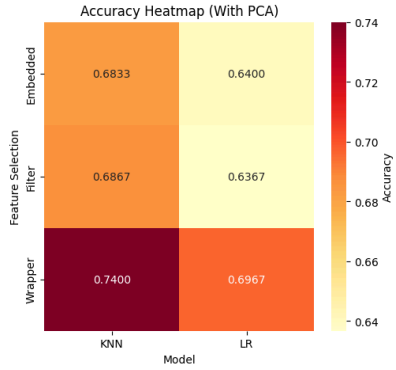

Wrapper Method PCA
Explained Variance Ratio
Principal Component

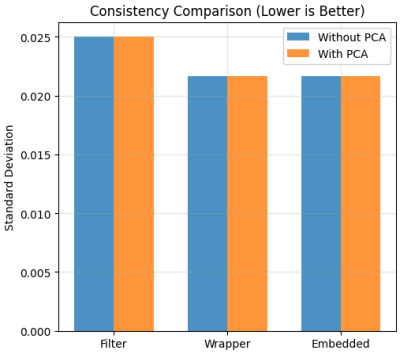

Embedded Method PCA
Explained Variance Ratio
Principal Component

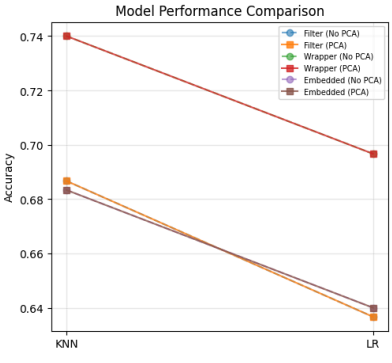

Accuracy: With vs Without PCA
Without PCA
With PCA
Accuracy
Filter KNN
Filter LR
Wrapper KNN
Wrapper LR
Embedded KNN
Embedded LR
Method-Model Combination


Performance by Feature Selection Method
KNN (No PCA)
KNN (PCA)
LR (No PCA)
LR (PCA)
Accuracy
Filter
Wrapper
Embedded


Accuracy Heatmap (Without PCA)
Feature Selection
Embedded
Filter
Wrapper
0.6833
0.6400
0.6867
0.6367
0.7400
0.6967
KNN
LR
Model
Accuracy


Accuracy Heatmap (With PCA)
Feature Selection
Embedded
Filter
Wrapper
0.6833
0.6400
0.6867
0.6367
0.7400
0.6967
KNN
LR
Model
Accuracy


Consistency Comparison (Lower is Better)
Without PCA
With PCA
Standard Deviation
Filter
Wrapper
Embedded


Model Performance Comparison
Filter (No PCA)
Filter (PCA)
Wrapper (No PCA)
Wrapper (PCA)
Embedded (No PCA)
Embedded (PCA)
Accuracy
KNN
LR